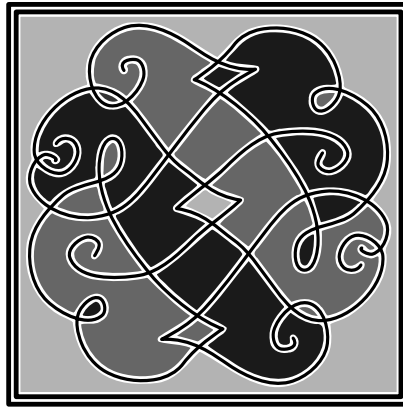


DYSON DEGREE APPRENTICESHIP (B.ENG.)
CONTROL THEORY & APPLICATIONS : WM213
TRIMESTER II 2019

REVISION : LAPLACE TRANSFORMS



1. INTRODUCTION TO LAPLACE TRANSFORMS
2. DERIVATIVES AND SHIFTING THEOREMS
3. SOLVING DIFFERENTIAL EQUATIONS

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Summary of Main Results

• **Definition of the Laplace Transform:** The Laplace transform $F(s) = \mathcal{L}[f]$ of a function $f \equiv f(t)$ is defined for $s > 0$ by

$$F(s) = \mathcal{L}[f] = \int_0^{\infty} e^{-st} f(t) dt.$$

One can also define a reverse transform denoted $\mathcal{L}^{-1}[F(s)] = f(t)$.

• **Solving Differential Equations by Laplace Transforms:**

1. Transform the given differential equation containing terms in $f(t)$ and its derivatives $f'(t)$, $f''(t)$, etc., to form a **subsidiary equation** comprised of terms in $F(s)$ only.
2. Solve the subsidiary equation for $F(s)$ using **algebraic manipulation**.
3. Finally, **reverse transform** the solution to find $f(t) = \mathcal{L}^{-1}[F(s)]$.

These steps are made easier with reference to tables of known Laplace transforms, such as those found in Sections 1 & 2 (see below).

• **Table of Laplace Transforms Found in Sections 1 & 2:**

Function	Transform	Function	Transform
$f(t)$	$F(s) = \int_0^{\infty} e^{-st} f(t) dt$	$e^{at} f(t)$	$F(s - a)$
1	$\frac{1}{s}$	$\cosh at$	$\frac{s}{s^2 - a^2}$
e^{at}	$\frac{1}{s - a}$	$\sinh at$	$\frac{a}{s^2 - a^2}$
t^n	$\frac{n!}{s^{n+1}}$	$t^n e^{at}$	$\frac{n!}{(s - a)^{n+1}}$
$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$	$e^{at} \cos \omega t$	$\frac{(s - a)}{(s - a)^2 + \omega^2}$
$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$	$e^{at} \sin \omega t$	$\frac{\omega}{(s - a)^2 + \omega^2}$
$H(t - a)$	$\frac{e^{-as}}{s}$	$f(t - a)H(t - a)$	$e^{-as} F(s)$
$f'(t)$	$sF(s) - f(0)$	$f''(t)$	$s^2 F(s) - sf(0) - f'(0)$

Section 1

CONTROL THEORY & APPLICATIONS

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Introduction to Laplace Transforms

Laplace transforms are integral transforms which can reduce the problem of solving differential equations to algebraic manipulation through a three step process:

1. A ‘difficult’ problem is transformed to a less difficult **subsidiary equation**.
2. The subsidiary equation is **solved using algebraic manipulation**.
3. The solution to the subsidiary equation is **reverse transformed** to yield the **solution** to the initial problem (usually with reference to transform tables).

This approach, through which problems in calculus are solved by manipulating transforms, is an example of what is called **operational calculus**.

Definition of the Laplace Transform

Let $f(t)$ be a function defined for all $t \geq 0$, then the **Laplace transform** of $f(t)$, denoted either $\mathcal{L}[f]$ or $F(s)$, where we shall assume $s > 0$ to be real, is defined

$$F(s) = \mathcal{L}[f] = \int_0^{\infty} e^{-st} f(t) dt, \quad \text{with } s > 0. \quad (1.1)$$

By ‘integrating out’ the t dependence, $F(s) = \mathcal{L}[f]$ is a function of s only; however, note that it is often convenient to write the transform of a function as $\mathcal{L}[f(t)]$.

- One may also define an **inverse transform**, denoted $\mathcal{L}^{-1}[F(s)]$, such that

$$f(t) = \mathcal{L}^{-1}[F(s)]. \quad (1.2)$$

- We shall use lowercase notation for $f(t)$ functions in ‘ t -space’, and uppercase notation $F(s)$ for their transform in ‘ s -space’.

Some Examples of Laplace Transforms

Example 1 Find the Laplace transform of unity, i.e., $f(t) = 1$:

$$\begin{aligned} F(s) = \mathcal{L}[1] &= \int_0^{\infty} e^{-st} dt = \left[-\frac{e^{-st}}{s} \right]_0^{\infty} \\ &= -\frac{1}{s} \left[\lim_{t \rightarrow \infty} e^{-st} - 1 \right] = -\frac{1}{s} [0 - 1], \quad \text{for } s > 0, \\ &= \frac{1}{s}, \quad \text{for } s > 0, \end{aligned} \quad (1.3)$$

Example 2 Find the Laplace transform of e^{at} , where a is constant:

$$\begin{aligned} F(s) = \mathcal{L}[e^{at}] &= \int_0^{\infty} e^{-st} e^{at} dt = \int_0^{\infty} e^{-(s-a)t} dt = \left[-\frac{e^{-(s-a)t}}{(s-a)} \right]_0^{\infty} \\ &= -\frac{1}{(s-a)} [0 - 1] = \frac{1}{(s-a)}, \quad \text{for } s > a. \end{aligned} \quad (1.4)$$

Note: Here the integral only converges when $s > a$, so $\mathcal{L}[e^{at}]$ only exists if $s > a$.

Example 3 Find the Laplace transform of t^n , where n is a positive integer:

$$\begin{aligned} F(s) = \mathcal{L}[t^n] &= \int_0^{\infty} e^{-st} t^n dt = \left[-\frac{t^n e^{-st}}{s} \right]_0^{\infty} + \frac{n}{s} \int_0^{\infty} e^{-st} t^{n-1} dt \\ &= \frac{n}{s} \times \mathcal{L}[t^{n-1}] \\ &= \frac{n}{s} \times \left(\frac{(n-1)}{s} \mathcal{L}[t^{n-2}] \right) \\ &= \frac{n}{s} \times \frac{(n-1)}{s} \times \frac{(n-2)}{s} \times \cdots \times \frac{1}{s} \mathcal{L}[1], \quad \text{for } s > 0, \end{aligned} \quad (1.5)$$

where we used integration by parts, i.e.,

$$\int_0^{\infty} u \frac{dv}{dt} dt = [uv]_0^{\infty} - \int_0^{\infty} v \frac{du}{dt} dt \quad (1.6)$$

with the functions $v = -e^{-st}/s$ and $u = t^n$. Hence, because $\mathcal{L}[1] = 1/s$ (see example 1), we have

$$\mathcal{L}[t^n] = \frac{n!}{s^{n+1}}, \quad \text{for } s > 0, \quad (1.7)$$

where $n! = n \times (n-1) \times (n-2) \times \cdots \times 1$ is ‘ n -factorial’.

Theorem 1 Linearity of Laplace Transforms: Laplace transforms are linear operations; that is, for any two functions $f(t)$ and $g(t)$ whose Laplace transforms exist, and for any two constants a and b , we have

$$\mathcal{L}[af(t) + bg(t)] = a\mathcal{L}[f(t)] + b\mathcal{L}[g(t)]. \quad (1.8)$$

Proof: By the definition in equation (1.1),

$$\begin{aligned} \mathcal{L}[af(t) + bg(t)] &= \int_0^\infty e^{-st}[af(t) + bg(t)]dt \\ &= a \int_0^\infty e^{-st}f(t)dt + b \int_0^\infty e^{-st}g(t)dt \\ &= a\mathcal{L}[f(t)] + b\mathcal{L}[g(t)]. \end{aligned} \quad (1.9)$$

Example 4 Find the Laplace transforms of $\sin \omega t$ and $\cos \omega t$, for constant ω : Euler's equation gives

$$e^{i\omega t} = (\cos \omega t + i \sin \omega t), \quad (1.10)$$

so that by theorem 1, and example 2 (i.e., using $\mathcal{L}[e^{at}] = \frac{1}{s-a}$ with $a = i\omega$) we have

$$\begin{aligned} \mathcal{L}[e^{i\omega t}] &= \mathcal{L}[\cos \omega t] + i\mathcal{L}[\sin \omega t] = \frac{1}{s - i\omega}, \quad \text{for } s > 0, \\ &= \frac{1}{s - i\omega} \cdot \left(\frac{s + i\omega}{s + i\omega} \right) \\ &= \frac{s}{s^2 + \omega^2} + \frac{i\omega}{s^2 + \omega^2}, \quad \text{for } s > 0. \end{aligned} \quad (1.11)$$

Hence, equating real and imaginary parts on the left and right-hand side,

$$\mathcal{L}[\cos \omega t] = \frac{s}{s^2 + \omega^2}, \quad \text{for } s > 0, \quad (1.12a)$$

$$\text{and } \mathcal{L}[\sin \omega t] = \frac{\omega}{s^2 + \omega^2}, \quad \text{for } s > 0. \quad (1.12b)$$

Exercise: Using the above method, find the Laplace transforms of $e^{at} \sin \omega t$ and $e^{at} \cos \omega t$, where a and ω are constant. [**Hint:** Consider the transform of $e^{at} e^{i\omega t}$.]

Example 5 Find the Laplace transforms of $\sinh at$ and $\cosh at$, for constant a : Since $\cosh at = \frac{1}{2}(e^{at} + e^{-at})$, by theorem 1 and example 2 we have

$$\begin{aligned}\mathcal{L}[\cosh at] &= \frac{1}{2} (\mathcal{L}[e^{at}] + \mathcal{L}[e^{-at}]) \\ &= \frac{1}{2} \left(\frac{1}{s-a} + \frac{1}{s+a} \right), \quad \text{for } s > a, \\ &= \frac{s}{s^2 - a^2}, \quad \text{for } s > a.\end{aligned}\tag{1.13}$$

Likewise, since $\sinh at = \frac{1}{2}(e^{at} - e^{-at})$, we find

$$\begin{aligned}\mathcal{L}[\sinh at] &= \frac{1}{2} (\mathcal{L}[e^{at}] - \mathcal{L}[e^{-at}]) \\ &= \frac{1}{2} \left(\frac{1}{s-a} - \frac{1}{s+a} \right), \quad \text{for } s > a, \\ &= \frac{a}{s^2 - a^2}, \quad \text{for } s > a.\end{aligned}\tag{1.14}$$

Gamma Function (Bonus)

For positive real numbers $x \in \mathbb{R}^+$, the gamma function $\Gamma(x)$ is defined

$$\boxed{\Gamma(x) \equiv \int_0^\infty \tau^{x-1} e^{-\tau} d\tau.}\tag{1.15}$$

Example 6 Find the Laplace transform of t^a , where a is a positive real number: Integration by parts (as in example 3, but with $v = -e^{-st}/s$ and $u = t^a$) yields

$$\begin{aligned}\mathcal{L}[t^a] &= \int_0^\infty e^{-st} t^a dt = \left[-\frac{t^a e^{-st}}{s} \right]_0^\infty + \int_0^\infty \frac{d}{dt}(t^a) \frac{e^{-st}}{s} dt \\ &= \frac{a}{s} \int_0^\infty t^{a-1} e^{-st} dt.\end{aligned}\tag{1.16}$$

Hence, applying the substitution $\tau = st$, with $d\tau = s dt$, and referring to definition (1.15), we have (for $s > 0$)

$$\mathcal{L}[t^a] = \frac{a}{s^{a+1}} \int_0^\infty \tau^{a-1} e^{-\tau} d\tau = \frac{a\Gamma(a)}{s^{a+1}}.\tag{1.17}$$

Exercise: Using integration by parts and the definition of the gamma function, show that $n\Gamma(n) = n!$ for integers $n > 0$. Hence derive $\mathcal{L}[t^n]$ from equation (1.17).

Section 2

Transforms of Derivatives and Shifting Theorems

Before we are ready to use Laplace transforms to solve ordinary differential equations (ODEs), we must discuss the transforms of differentials and also find those of a few more functions. In this section we consider (with constant a):

- The first shifting theorem: (s -shifting) replacing s by $(s - a)$ in $F(s)$.
- The second shifting theorem: (t -shifting) replacing t by $(t - a)$ in $f(t)$.
- Differentiation of $f(t)$ as ‘roughly’ representing multiplication of $F(s)$ by s .

For the purpose of these revision notes, the shifting theorems may be considered optional. All the Laplace transforms calculated so far are gathered for reference in table 2.1 at the end.

Theorem 2 First Shifting Theorem (optional): (also called s -shifting)
If $F(s) = \mathcal{L}[f(t)]$ is the Laplace transform of $f(t)$, then

$$\mathcal{L}[e^{at}f(t)] = F(s - a) \quad (2.1)$$

Proof: By the definition of the Laplace transform in equation (1.1),

$$\begin{aligned} F(s) = \int_0^\infty e^{-st}f(t)dt &\Leftrightarrow F(s - a) = \int_0^\infty e^{-(s-a)t}f(t)dt \\ &= \int_0^\infty e^{-st}[e^{at}f(t)]dt = \mathcal{L}[e^{at}f(t)] . \end{aligned} \quad (2.2)$$

Corollary: From equation (2.2) above, if $F(s - a) = \mathcal{L}[e^{at}f(t)]$ is the Laplace transform of $e^{at}f(t)$, then $F(s)$ is the Laplace transform of $f(t)$.

Example 7 Find the Laplace transforms of $e^{at} \sin \omega t$ and $e^{at} \cos \omega t$:

By s -shifting the transforms from example 4 using theorem 2, that is, making the transformation $s \rightarrow (s - a)$, we have

$$\mathcal{L}[e^{at} \cos \omega t] = \frac{(s - a)}{(s - a)^2 + \omega^2}, \quad \text{for } s > a, \quad (2.3a)$$

$$\text{and } \mathcal{L}[e^{at} \sin \omega t] = \frac{\omega}{(s - a)^2 + \omega^2}, \quad \text{for } s > a. \quad (2.3b)$$

[This should agree with your answer to the exercise in example 4 of Lecture 1.]

Example 8 Find the Laplace transform of $f(t) = t^n e^{at}$:

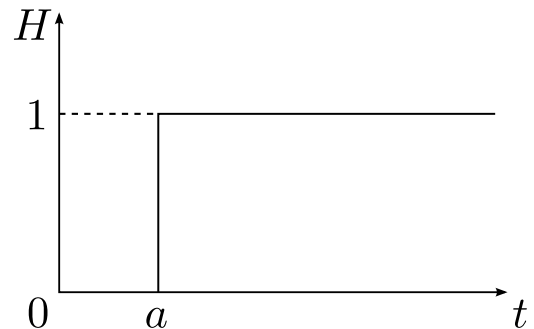
By theorem 2 and example 3 with $F(s) = \mathcal{L}[t^n]$ for $s > 0$ we have

$$\mathcal{L}[t^n e^{at}] = F(s - a) = \frac{n!}{(s - a)^{n+1}}, \quad \text{for } s > a. \quad (2.4)$$

Heaviside Function

The Heaviside function $H(t - a)$ is defined for constant $a \geq 0$ by

$$H(t - a) = \begin{cases} 0, & \text{for } t < a \\ 1, & \text{for } t > a, \end{cases} \quad (2.5)$$



as shown in figure 2.1; it is discontinuous at a , where we leave it undefined.

Figure 2.1: Heaviside Function.

Example 9 Find the Laplace transform of the Heaviside function $H(t - a)$:

Since $H(t - a)$ is zero on the interval $t \in [0, a)$, this part does not contribute to the transform integral and we can change the limits from $t : 0 \rightarrow \infty$ to $t : a \rightarrow \infty$. Thus, for $s > 0$ we obtain

$$\begin{aligned} \mathcal{L}[H(t - a)] &= \int_0^\infty e^{-st} H(t - a) dt = \int_a^\infty e^{-st} dt \\ &= -\frac{1}{s} [e^{-st}]_a^\infty \\ &= \frac{e^{-as}}{s}. \end{aligned} \quad (2.6)$$

Theorem 3 Second Shifting Theorem (optional): (t -shifting) If $F(s)$ is the Laplace transform of $f(t)$, and $H(t - a)$ is the Heaviside function, then

$$\mathcal{L}[f(t - a)H(t - a)] = e^{-as}F(s). \quad (2.7)$$

Proof: Using the dummy integration variable $\tau = t + a$, we have

$$\begin{aligned} e^{-as}F(s) &= e^{-as} \int_0^\infty e^{-st} f(t) dt = \int_0^\infty e^{-s(a+t)} f(t) dt = \int_a^\infty e^{-s\tau} f(\tau - a) d\tau \\ &= \int_0^\infty e^{-s\tau} [f(\tau - a)H(\tau - a)] d\tau = \mathcal{L}[f(t - a)H(t - a)]. \end{aligned} \quad (2.8)$$

With $f(t) = 1$ and the Laplace transform of unity, $\mathcal{L}[1] = 1/s$ (see example 1), this theorem yields $\mathcal{L}[H(t - a)] = e^{-as}/s$, as expected from equation (2.6).

- Replacing t by $(t - a)$ in $f(t)$ ‘roughly’ equates to multiplying $F(s)$ by e^{-as} .

Transforms of Derivatives

Laplace transforms of first and second-order derivatives may be found as follows.

Example 10 Find the Laplace transform of $f'(t) \equiv \frac{df}{dt}$ and $f''(t) \equiv \frac{d^2f}{dt^2}$:

Assuming that $f(t)$ is a suitably ‘well behaved’ function, such that its transform and those of its derivatives exist, then using integration by parts we have

$$\begin{aligned} \mathcal{L}[f'] &= \int_0^\infty e^{-st} \frac{df}{dt} dt = [f(t)e^{-st}]_0^\infty + s \int_0^\infty e^{-st} f(t) dt \\ &= \left[\lim_{t \rightarrow \infty} f(t)e^{-st} - f(0) \right] + s\mathcal{L}[f] \\ &= s\mathcal{L}[f] - f(0). \end{aligned} \quad (2.9)$$

Similarly, from this result for $\mathcal{L}[f']$, we find

$$\begin{aligned} \mathcal{L}[f''] &= \int_0^\infty e^{-st} \frac{d^2f}{dt^2} dt = \left[\frac{df}{dt} e^{-st} \right]_0^\infty + s \int_0^\infty e^{-st} \frac{df}{dt} dt \\ &= s\mathcal{L}[f'] - f'(0) \\ &= s^2\mathcal{L}[f] - sf(0) - f'(0). \end{aligned} \quad (2.10)$$

Theorem 4 Laplace transform of a derivative of order n (optional):

As in example 10, let us assume that $f(t)$ and its derivatives are suitably ‘well behaved’ so that each of our steps is meaningful. Thus, taking $f^n \equiv \frac{d^n f}{dt^n}$, integration by parts followed by re-substitution for the definition of the Laplace Transform gives

$$\begin{aligned}\mathcal{L}[f^n] &= \int_0^\infty e^{-st} \frac{d^n f}{dt^n} dt = s \int_0^\infty e^{-st} \frac{d^{n-1} f}{dt^{n-1}} dt + \left[\frac{d^{n-1} f}{dt^{n-1}} e^{-st} \right]_0^\infty \\ &= s\mathcal{L}[f^{n-1}] - f^{n-1}(0) \\ &= s \{ s\mathcal{L}[f^{n-2}] - f^{n-2}(0) \} - f^{n-1}(0), \quad \text{and so forth.} \quad (2.11)\end{aligned}$$

Hence, with $\mathcal{L}[f^1] = s\mathcal{L}[f] - f(0)$ (see example 10), by mathematical induction,

$$\mathcal{L}[f^n] = s^n \mathcal{L}[f] - s^{n-1} f(0) - s^{n-2} f^1(0) - \dots - s f^{n-2}(0) - f^{n-1}(0). \quad (2.12)$$

Summary Table of Laplace Transforms

Function	Transform	Function	Transform
$f(t)$	$F(s) = \int_0^\infty e^{-st} f(t) dt$	$e^{at} f(t)$	$F(s-a)$
1	$\frac{1}{s}$	$\cosh at$	$\frac{s}{s^2 - a^2}$
e^{at}	$\frac{1}{s-a}$	$\sinh at$	$\frac{a}{s^2 - a^2}$
t^n	$\frac{n!}{s^{n+1}}$	$t^n e^{at}$	$\frac{n!}{(s-a)^{n+1}}$
$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$	$e^{at} \cos \omega t$	$\frac{(s-a)}{(s-a)^2 + \omega^2}$
$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$	$e^{at} \sin \omega t$	$\frac{\omega}{(s-a)^2 + \omega^2}$
$H(t-a)$	$\frac{e^{-as}}{s}$	$f(t-a)H(t-a)$	$e^{-as} F(s)$
$f'(t)$	$sF(s) - f(0)$	$f''(t)$	$s^2 F(s) - sf(0) - f'(0)$

Table 2.1: Summary of Laplace transforms calculated in sections 1 and 2. Remember: $F(s) = \mathcal{L}[f]$.

Section 3

CONTROL THEORY & APPLICATIONS

J. J. BISSELL, TRIMESTER II 2019

Solving Differential Equations

Using the Laplace transforms in table 2.1 we are now in a position to solve some basic differential equations. Our method (see lecture 1) involves three steps:

- **Step 1:** Transform the given differential equation containing terms in $f(t)$, $f'(t)$, $f''(t)$, etc., to form a **subsidiary equation** with terms in $F(s)$ only.
- **Step 2:** Solve the subsidiary equation for F using **algebraic manipulation**.
- **Step 3:** **Reverse transform** the solution to find $f(t) = \mathcal{L}^{-1}[F(s)]$.

In this lecture we illustrate the method with a few examples, using lowercase for a function $f(t)$ in t -space, and uppercase for its transform $F(s)$ in s -space.

Solving First Order Ordinary Differential Equations

Example 11 Using the standard notation $\mathcal{L}[y(t)] = Y(s)$, solve the differential equation $y' + ky = 0$, for constant $k > 0$, subject to the initial condition $y(0) = 2$:

- **Step 1:** Transform the problem term by term using table 2.1 (cf. theorem 1):

$$y' + ky = 0 \quad \Rightarrow \quad \mathcal{L}[y'] + k\mathcal{L}[y] = \mathcal{L}[0] \quad \Rightarrow \quad sY(s) - y(0) + kY(s) = 0. \quad (3.1)$$

- **Step 2:** Solve the subsidiary equation:

$$sY(s) - y(0) + kY(s) = 0 \quad \Rightarrow \quad Y(s) = \frac{2}{s+k}. \quad (3.2)$$

- **Step 3:** Reverse transform $Y(s)$ using table 2.1 to give the solution:

$$y(t) = \mathcal{L}^{-1}[Y(s)] = 2\mathcal{L}^{-1}\left[\frac{1}{s+k}\right] = 2e^{-kt}, \quad (3.3)$$

where we used $\mathcal{L}^{-1}\left[\frac{1}{s-a}\right] = e^{at}$ with $a = -k$. Naturally, one can always check this solution by substituting it into the governing equation and initial condition.

Laplace Transform Method as Moving Between Domains

When solving models of physical systems, one may think of the Laplace transform method in terms of transferring between the **time domain** (or t -space) where the problem is defined, and the **transform domain** (or s -space) where we work with the problem's transform. Such an approach, through which problems in calculus are solved by manipulating transforms, is called **operational calculus**.

Let us illustrate this idea with a familiar problem. Considering charging a capacitor C in series with a resistor R by connecting it to a battery of voltage V as in the RC -circuit diagram in figure 3.1.

Denoting the current flowing through the circuit as $I(t) = dq/dt$, where dq/dt is the rate-of-change of charge $q(t)$ on the capacitor at time t , then the voltage drop across the resistor V_R is simply $V_R(t) = I(t)R = (dq/dt)R$, while the potential difference between the capacitor's plates is $V_C(t) = q(t)/C$.

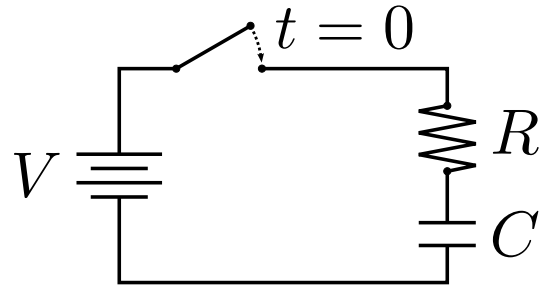


Figure 3.1: Capacitor C in series with a resistor R . The switch is closed at $t = 0$.

If a connection is made by closing the switch at time $t = 0$, then we can model the voltage V_S supplied by the battery using the **Heaviside function** $H(t)$, that is, $V_S(t) = VH(t)$.¹ Hence, equating appropriate voltages we have $V_S(t) = V_R(t) + V_C(t)$, and thus

$$\frac{dq(t)}{dt} + \frac{1}{RC}q(t) = \frac{V}{R}H(t) \quad (3.4)$$

as the differential equation governing the charge on the capacitor $q(t)$ at time t . Assuming the capacitor to be uncharged at $t = 0$, i.e., the **initial condition** $q(t) = 0$, we now solve this equation by the method of Laplace transforms. As usual, we adopt the standard lowercase convention for $q(t)$ in t -space, and the uppercase convention for its transform $Q(s) = \mathcal{L}[q(t)]$ in s -space.

Essentially, Laplace transforms help us to solve differential equations formulated in the **time domain** (t -space) by allowing us to work in a **transform domain**

¹Recall that the Heaviside function is defined such that $H(t) = 0$ for $t < 0$, and $H(t) = 1$ for $t > 0$, hence its suitability for describing the switch connection: $t < 0$ gives 'off', and $t > 0$ gives 'on' (see figure 2.1, lecture 2).

(s -space) where solutions may be found using **algebraic manipulation**. The following table narrates this process of moving between domains for the capacitor charging problem defined in equation (3.4); notice in particular how **Laplace transforms** ($\mathcal{L}$) move us from t -space to s -space, while **inverse Laplace transforms** (or reverse transforms, $\mathcal{L}^{-1}$) move us from s -space to t -space:

Time Domain (t -space)	$\rightleftharpoons$	Transform Domain (s -space)
Initial Problem The ordinary differential equation which models charging a capacitor (and that we want to solve) is $q'(t) + \frac{1}{RC}q(t) = \frac{V}{R}H(t).$ This equation has terms in both $q(t)$ and its derivative $q'(t)$; crucially, after transformation, both of these components yield terms in $Q(s)$ only. An uncharged capacitor at $t = 0$ gives the initial condition $q(0) = 0$. ($\downarrow$) Step 3 : Reverse Transform $Q(s)$ to Obtain $q(t) = \mathcal{L}^{-1}[Q(s)]$ Finally, by reverse transforming $Q(s)$, we switch back to t-space and obtain the required solution $q(t) = \mathcal{L}^{-1}[Q(s)]$, that is, $q = CV \left(\mathcal{L}^{-1} \left[\frac{1}{s} \right] - \mathcal{L}^{-1} \left[\frac{1}{s+1/RC} \right] \right)$ $= CV \left(1 - e^{-t/RC} \right).$ Note: Here we used our Laplace transform table in reverse to give $\mathcal{L}^{-1} \left[\frac{1}{s} \right] = 1$ and $\mathcal{L}^{-1} \left[\frac{1}{s-a} \right] = e^{at}$, with $a = -1/RC$.	$\mathcal{L}$ $\longrightarrow$	Step 1 : Transform Problem By Laplace transforming the problem we switch to s-space, thus: $\mathcal{L}[q'] + \frac{1}{RC}\mathcal{L}[q] = \frac{V}{R}\mathcal{L}[H(t)],$ a form which may be expressed with reference to tables of known transforms as the subsidiary equation $sQ(s) - q(0) + \frac{1}{RC}Q(s) = \frac{V}{R} \frac{1}{s}.$ Notice that this equation has unknowns in $Q(s) = \mathcal{L}[q(t)]$ only. $\downarrow$ Step 2 : Solve the Subsidiary Equation for $Q(s)$ Algebraically With initial condition $q(0) = 0$, the subsidiary equation may be solved using algebraic manipulation, <i>viz</i> $Q(s) = \frac{V/R}{s(s+1/RC)}.$ Hence, separating the terms by the method of partial fractions, we have $Q(s) = CV \left(\frac{1}{s} - \frac{1}{s+1/RC} \right)$ for use with the tables.
	$\mathcal{L}^{-1}$ $\longleftarrow$	

Solving Second Order Ordinary Differential Equations

Example 12 Solve the differential equation $y'' + 4y' - 5y = 0$ subject to the initial conditions $y'(0) = 2$ and $y(0) = 1$:

• **Step 1:** We transform the problem term by term using table 2.1:

$$y'' + 4y' - 5y = 0 \quad \Rightarrow \quad s^2Y(s) - sy(0) - y'(0) + 4sY(s) - 4y(0) - 5Y(s) = 0. \quad (3.5)$$

• **Step 2:** We solve the subsidiary equation noting the initial conditions:

$$s^2Y(s) - s - 2 + 4sY(s) - 4y - 5Y(s) = 0 \quad \Rightarrow \quad Y(s) = \frac{(s+6)}{(s^2+4s-5)}. \quad (3.6)$$

This does not yet look like any of our known Laplace transforms - we need an expression comprising separate terms, so factorise the denominator:

$$Y(s) = \frac{(s+6)}{(s^2+4s-5)} = \frac{(s+5)+1}{(s+5)(s-1)} = \frac{1}{(s-1)} + \frac{1}{(s+5)(s-1)}. \quad (3.7)$$

To reverse transform using table 2.1, we *still* need to separate the final term which we can do using standard methods for partial fractions. Indeed, with constant A and B we have

$$\frac{1}{(s+5)(s-1)} = \frac{A}{(s+5)} + \frac{B}{(s-1)} \quad \Rightarrow \quad 1 = A(s-1) + B(s+5). \quad (3.8)$$

Taking $s = -5$ and $s = 1$ yields $A = -1/6$ and $B = 1/6$ respectively, so that combining equations (3.7) and (3.8) we have

$$Y(s) = \frac{1}{(s-1)} - \frac{1}{6(s+5)} + \frac{1}{6(s-1)} = \frac{7}{6(s-1)} - \frac{1}{6(s+5)}. \quad (3.9)$$

• **Step 3:** Finally, we reverse transform $Y(s)$ using table 2.1 to give the solution:

$$y(t) = \mathcal{L}^{-1}[Y(s)] = \frac{7}{6}\mathcal{L}^{-1}\left[\frac{1}{(s-1)}\right] + \frac{1}{6}\mathcal{L}^{-1}\left[\frac{1}{(s+5)}\right] \quad (3.10)$$

$$= \frac{7}{6}e^t - \frac{1}{6}e^{-5t} \quad (3.11)$$

As in example 11, one can readily check this solution by substituting it into the governing differential equation and initial conditions.